

□ Find the power series solution of the differential equation $\frac{d^2y}{dx^2} + x \frac{dy}{dx} + (Rx+2)y = 0$

Taking $R=9$, the differential equation is,

$$\frac{d^2y}{dx^2} + x \frac{dy}{dx} + (9x+2)y = 0$$

Since, $a_0(x)=1$;

This power series must be solved for ordinary point

Now,

$$y = \sum_{n=0}^{\infty} a_n x^n$$

$$\therefore y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$\therefore y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$$

So, the equation becomes,

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + x \sum_{n=1}^{\infty} n a_n x^{n-1} + 9x \sum_{n=0}^{\infty} a_n x^n + 2 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=1}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} 9 a_n x^{n+1} + \sum_{n=0}^{\infty} 2 a_n x^n = 0$$

By equating the coefficients of x^n ,

$$(n+2)(n+1) a_{n+2} + (n+2) a_n + 9 a_{n-1} = 0$$

$$\Rightarrow a_{n+2} = \frac{-(n+2) a_n - 9 a_{n-1}}{(n+2)(n+1)}$$

For, $n=0$;

$$a_2 = \frac{-(2)a_0 - 0}{(2)(1)} = -a_0$$

For, $n=1$;

$$a_3 = \frac{-(3)a_1 - 9a_0}{(3)(2)} = -\frac{(a_1 + 3a_0)}{2}$$

For, $n=2$;

$$a_4 = \frac{-(4)a_2 - 9a_1}{(4)(3)} = \frac{-4(-a_0) - 9a_1}{12} = \frac{4a_0 - 9a_1}{12}$$

Final series solution,

$$y = a_0(1 - x^2 - \frac{3}{2}x^3 + \frac{1}{3}x^4 + \dots) + a_1(x - \frac{1}{2}x^3 - \frac{3}{4}x^4 + \dots)$$

Ans.

□ Find the power series solution of the differential equation $(2x^2-3)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 9y = 0$; $y(0) = -1$, $y'(0) = 5$

Taking $R=9$, the differential equation is,

$$(2x^2-3)y'' - 2xy' + 9y = 0$$

Since, $a_0(x) = 2x^2 - 3$

$$\Rightarrow a_0(0) = -3 \neq 0;$$

This power series must be solved for ordinary point

Now,

$$y = \sum_{n=0}^{\infty} a_n x^n$$

$$\therefore y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$\therefore y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

So, the equation becomes,

$$(2x^2-3) \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - 2x \sum_{n=1}^{\infty} n a_n x^{n-1} + 9 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{n=2}^{\infty} 2n(n-1) a_n x^n - \sum_{n=2}^{\infty} 3n(n-1) a_n x^{n-2} - \sum_{n=1}^{\infty} 2n a_n x^n + \sum_{n=0}^{\infty} 9 a_n x^n = 0$$

$$\Rightarrow \sum_{n=2}^{\infty} 2n(n-1) a_n x^n - \sum_{n=0}^{\infty} 3(n+2)(n+1) a_{n+2} x^n - \sum_{n=1}^{\infty} 2n a_n x^n + \sum_{n=0}^{\infty} 9 a_n x^n = 0$$

By equating the coefficients of x^n ,

$$-3(n+2)(n+1) a_{n+2} + (2n(n-1) - 2n + 9) a_n = 0$$

$$\Rightarrow a_{n+2} = \frac{(2n^2 - 4n + 9) a_n}{3(n+2)(n+1)}$$

For, $n=0$;

$$a_2 = \frac{(2(0)^2 - 4(0) + 9)a_0}{3(2)(1)} = \frac{9a_0}{2} = \frac{3}{2}(-1) = -3/2$$

For, $n=1$;

$$a_3 = \frac{(2(1)^2 - 4(1) + 9)a_1}{3(3)(2)} = \frac{35}{18}$$

For, $n=2$;

$$a_4 = \frac{(2(2)^2 - 4(2) + 9)a_2}{3(4)(3)} = \frac{(8-8+9)a_2}{36} = \frac{9}{36}(-3/2) = -3/8$$

Final Series Solution,

$$y = -1 + 5x - \frac{3}{2}x^2 + \frac{35}{18}x^3 - \frac{3}{8}x^4 + \dots$$

Ans.

□ Using Method of Frobenius find solution near $x=0$ of the differential equation $x \frac{d^2 y}{dx^2} - \frac{9}{2} \frac{dy}{dx} - y = 0$

Taking $R=9$, the differential equation is,

$$x \frac{d^2 y}{dx^2} - \frac{9}{2} \frac{dy}{dx} - y = 0$$

Since, $a_0(x) = x$

$$\Rightarrow a_0(0) = 0;$$

Hence, $x=0$ is a singular point

Frobenius Series;

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

$$\therefore y' = \sum_{n=0}^{\infty} a_n (n+r) x^{n+r-1}$$

$$\therefore y'' = \sum_{n=0}^{\infty} a_n (n+r)(n+r-1) x^{n+r-2}$$

So, the equation becomes,

$$x \sum_{n=0}^{\infty} a_n (n+r)(n+r-1) x^{n+r-2} - \frac{9}{2} \sum_{n=0}^{\infty} a_n (n+r) x^{n+r-1} - \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} a_n (n+r)(n+r-1) x^{n+r-1} - \frac{9}{2} \sum_{n=0}^{\infty} a_n (n+r) x^{n+r-1} - \sum_{n=0}^{\infty} a_n x^{n+r} = 0$$

Here,

The lowest power of x is x^{r-1} ($\because n=0$). Setting its coefficient to zero;

$$a_0 (r(r-1) - \frac{9}{2}r) = 0$$

$$\Rightarrow r^2 - r - \frac{9}{2}r = 0 \quad (\because a_0 \neq 0)$$

$$\Rightarrow r_1 = 11/2, r_2 = 0$$

For a general n , aligning all terms to x^{n+r-1} ,

$$((n+r)(n+r-1) - \frac{9}{2}(n+r)) a_n - a_{n-1} = 0$$

$$\Rightarrow (n+r)(n+r-1 - \frac{9}{2}) a_n = a_{n-1}$$

$$\Rightarrow a_n = \frac{a_{n-1}}{(n+r)(n+r - \frac{11}{2})}$$

For, $r = \frac{11}{2}$;

$$a_n = \frac{a_{n-1}}{(n+\frac{11}{2})(n)}$$

For, $n=1$;

$$a_1 = \frac{a_0}{(1+\frac{11}{2})(1)} = \frac{2a_0}{13}$$

For, $n=2$;

$$a_2 = \frac{a_1}{(2+\frac{11}{2})(2)} = \frac{2}{195} a_0$$

$$\therefore y_1 = a_0 x^{11/2} \left(1 + \frac{2}{13} x + \frac{2}{195} x^2 + \dots \right)$$

For, $r=0$

$$a_n = \frac{a_{n-1}}{n(n-\frac{11}{2})}$$

For, $n=1$;

$$a_1 = \frac{a_0}{1(1-\frac{11}{2})} = -\frac{2}{9} a_0$$

For, $n=2$;

$$a_2 = \frac{a_1}{2(2-\frac{11}{2})} = \frac{2}{63} a_0$$

For, $n=3$;

$$a_3 = \frac{a_2}{3(3-\frac{11}{2})} = -\frac{4}{945} a_0$$

$$\therefore y_2 = a_0 \left(1 - \frac{2}{9} x + \frac{2}{63} x^2 - \frac{4}{945} x^3 + \dots \right)$$

$$\therefore y = y_1 + y_2 = A x^{11/2} \left(1 + \frac{2}{13} x + \frac{2}{195} x^2 + \dots \right) + B \left(1 - \frac{2}{9} x + \frac{2}{63} x^2 - \frac{4}{945} x^3 + \dots \right)$$

Ans.